Discover Chemistry

Brief Communication

Comprehensive screening of per- and polyfluoroalkyl substances in consumer products using pyrolysis GC–MS

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Abstract
This paper presents a screening method for diverse classes of per- and polyfluoroalkyl substances (PFAS) in consumer products using a Pyrolysis Gas Chromatography Mass Spectrometer (Py-GC–MS). During the pyrolysis process, polar functional groups (carboxylic or sulfonic) are thermally dissociated, allowing the non-polar perfluoroalkyl moieties to elute through the GC column. We successfully detected various classes of PFAS, from non-polymeric forms of different chain lengths to polymeric forms, within 30 min, achieving detection limits below 1/100 of the typical intentional use concentrations of PFAS in consumer products. Therefore, the screening method can detect even low levels of intentionally added PFAS within a comfortable margin.

1 Introduction

Per- and polyfluoroalkyl substances (PFAS), often termed "forever chemicals," have been recognized as environmental contaminants across various regions due to their persistence, bioaccumulation, and potential adverse impacts on human health and the environment [1, 2]. Mass production of PFAS began in the 1950s by US chemical companies, and the use of these substances has grown since the 1970s across diverse applications, including firefighting foams, water-repellent fabrics, fast-food containers, photolithography materials, and flame retardants [3–5]. From production to disposal, PFAS are released into the environment at every stage of a product's lifecycle, accumulating in organisms [6, 7]. The Stockholm Convention on Persistent Organic Pollutants (POPs) has listed certain PFAS, such as perfluoro-octane sulfonic acid (PFOS) and perfluoro-octanoic acid (PFOA), for global elimination, prompting many countries to impose restrictions [8]. For instance, the European Union has not only banned specific PFAS like PFOS and PFOA but is also considering a universal restriction that would regulate PFAS as a class due to their shared properties [9, 10]. Meanwhile, the United States (U.S.) Environmental Protection Agency, along with several U.S. states, has recently introduced new PFAS regulations, with additional states expected to introduce new laws in the near future [11–13]. These U.S. regulations aim to cover a comprehensive range of PFAS in consumer products. Overall, global regulatory trends are shifting towards phasing out or significantly restricting PFAS through both substance-specific and class-based approaches [14]. Comprehensive PFAS regulation limits the intentional use of PFAS as a precautionary measure, prevents the use of regrettable substitutes, and facilitates transition to non-PFAS alternatives. To limit the intentional use of PFAS, an LOD below 10 mg/kg (approximately

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1/100 of the typical intentional use concentrations of 500—1,000 mg/kg) can provide a sufficient margin for detecting intentionally added PFAS in consumer products [15].

In this regard, analytical methods suitable for both substance-specific and comprehensive class-based bans are essential to adequately address PFAS regulations. Typically, organic compounds are analyzed using either gas chromatography (GC) or liquid chromatography (LC) techniques. The GC method, while effective, is limited in its ability to analyze compounds with low volatility. Conversely, the LC method does not face these limitations and is preferred for analyzing non-volatile and semi-volatile PFAS compounds. However, challenges persist for some PFAS, which lack well-defined structures and established toxicity endpoints, complicating their identification via LC–MS or LC–MS/MS. Additionally, propagating standard substances for all PFAS would be prohibitively expensive and unrealistic. Consequently, it is impractical to analyze numerous PFAS individually, necessitating the development of a new approach for collective analysis.

In response to these challenges, combustion ion chromatography (CIC) has been proposed as a means to collectively manage PFAS [16]. However, CIC can only measure the total amount of thermally decomposed fluorine and cannot distinguish between inorganic and organic sources, prompting the need for more specific methods to measure all types of PFAS exclusively. Aside from LC–MS and CIC, there are limited alternatives. Notably, some non-volatile compounds can be analyzed in conventional GC by creating volatile derivatives, but this process introduces an additional step, increasing the potential for errors and necessitating further method validation [17, 18]. Consequently, it is often considered more effective to analyze non-volatile compounds using other methods. Another possibility is the pyrolysis GC method. During pyrolysis, samples are rapidly degraded at high temperatures. If the polar functional groups can be thermally dissociated, the remaining non-polar perfluoroalkyl moiety can be easily eluted through the GC column, making the fragment ions detectable by GC–MS. This method's advantage is its ability to extract samples thermally, eliminating the need for solvent extraction. Additionally, a recent study demonstrated PFAS screening using a two-step approach combining CIC and Py-GC–MS [19]. Following qualitative identification by Py-GC–MS, total fluorine was quantitatively determined using CIC, with a detection limit set to meet a threshold of 50 mg F kg^{-1}. In short, CIC is useful for detecting total fluorine. GC–MS is reliable for detecting volatile PFAS, while LC–MS outperforms GC–MS for specific non-volatile and semi-volatile PFAS. Additionally, Pyrolysis GC–MS provides distinct advantages in detecting diverse class PFAS without requiring complex sample preparation.

The purpose of this study is to examine analytical methods for PFAS as regulations expand from substance-specific bans to comprehensive class-based restrictions and to confirm their effectiveness in addressing these comprehensive regulations. The goal is to improve the speed and simplicity of comprehensive PFAS analysis. Herein, a comprehensive PFAS analysis without the CIC step, using Py–GC–MS alone, was explored by targeting the distinct fragment ions of specific pyrolysis products.

2 Materials and method

2.1 Materials

Reagents used for method verification include Perfluorobutanoic acid (PFBA, C_4), Perfluorohexanoic acid (PFHxA, C_6), Perfluorooctanoic acid (PFOA, C_8), Perfluorodecanoic acid (PFDA, C_{10}), Perfluorododecanoic acid (PFDODA, C_{12}), Perfluorotetradecanoic acid (PFTEDA, C_{14}) Perfluorohexadecanoic acid (PFHxDA, C_{16}), Perfluorooctadecanoic acid (PFODA, C_{18}), Perfluorohexane sulfonic acid (PFHxS, C_6), and Perfluorooctane sulfonic acid (PFOS, C_8). Here, PFBA, PFOA, PFDA, PFHxS, and PFOS were procured from Tokyo Chemical Industry Co., Ltd, while PFDoDA, PFTeDA, PFHxDA, and PFODA were sourced from Wellington Laboratories. These reagents were diluted with methanol to create standard solutions as needed. The PFAS standard materials for method validation were mainly chosen from perfluoroalkyl carboxylic acids (PFCAs) and perfluoroalkane sulfonic acids (PFSAs), which are currently or likely to be restricted by the Stockholm Convention on POPs regulation. Additional PFAS standards, including Fluorotelomer alcohols (FTOHs) and other volatile analytical standards, will be prepared in future studies. In addition to these standards, real samples of PFAS-free polyester (catalog number: GF14511215; Sigma-Aldrich Co. LLC), total fluorine positive (polyester, polycarbonate, polyimide), and Perfluorobutansulfonic acid (PFBS) positive (polyester, polycarbonate, phenol formaldehyde novolac) were prepared for verification and comparison with the CIC and/or LC–MS/MS results. To construct a calibration curve, a specific quantity of pre-adjusted standard solution was added to align with a sample weight of 0.5 mg. For instance, for a calibration point

at 1,000 µg/g, 500 ng of analyte (equivalent to 500 ng/0.5 mg = 1,000 µg/g) was added to the sample cup. Additionally, commercially available PTFE seal tape (NICHIAS Corporation) was used as a reference for polymeric PFAS.

2.2 Method

Samples were analyzed using a GC–MS system (GCMS-QP2010, Shimadzu Corp.) equipped with a pyrolysis unit (EGA/PY-3030D, Frontier Lab) mounted atop the GC inlet. Figure 1 is a schematic of the Py-GC–MS system and the hypothesized mechanism of PFAS fragmentation.

Approximately 0.5 mg of either cut or powdered samples were placed in sample cups (Eco-Cup LF, Frontier Lab), then introduced to a pyrolysis furnace heated to 700 °C for analysis by the GC–MS unit. Measurements were performed on a 30 m GC column (I.D. 0.25 mm; film thickness: 0.25 µm; Ultra ALLOY-5; 95% methyl; 5% phenyl polysiloxane; Frontier Lab). During each GC run, the column temperature was held at 50 ℃ for 2 min, then increased from 50 ℃ to 290 ℃ at 20 ℃/min, and finally held at 290 °C for 6 min. The injection mode was set to split with a ratio of 1:50, and the carrier gas was helium at a constant linear velocity of 40.6 cm/s. Mass analysis was conducted in selective ion monitoring (SIM) and scan modes. PFAS chromatographic peaks typically contain abundant fragment ions with mass-to-charge ratios of 69 and 131 represented by CF_3^+ and $C_3F_5^+$, respectively. These characteristic fragment ions commonly appear in the mass spectra of PFAS. Herein, $C_3F_5^+$ (*m/z* 131) was selected as the quantification ion because it is little interfered by the sample matrices, whereas CF_3^+ (*m/z* 69) was selected as the confirmation ion.

3 Results and discussion

Given the relatively weak bond between the polar functional group (carboxylic or sulfonic) and the alkyl chain, in contrast to the strong C-F bond, dissociation of the polar functional groups is presumed to dominate the thermal degradation process, making PFAS detectable post-pyrolysis by GC–MS. Figures 2a and 2b display typical total ion current (TIC) chromatograms derived from PFCAs and two PFSAs with varying chain lengths, each at a concentration of 1,000 mg/kg. Additionally, a result from a polymeric PFAS (polytetrafluoroethylene; PTFE) is presented in Fig. 2(c).

As the alkyl chain lengthens from C4 to C18, the PFCAs exhibit slower elution through the GC column, resulting in the division of signals into two distinct chromatographic peaks. Data acquisition in SIM mode was conducted concurrently

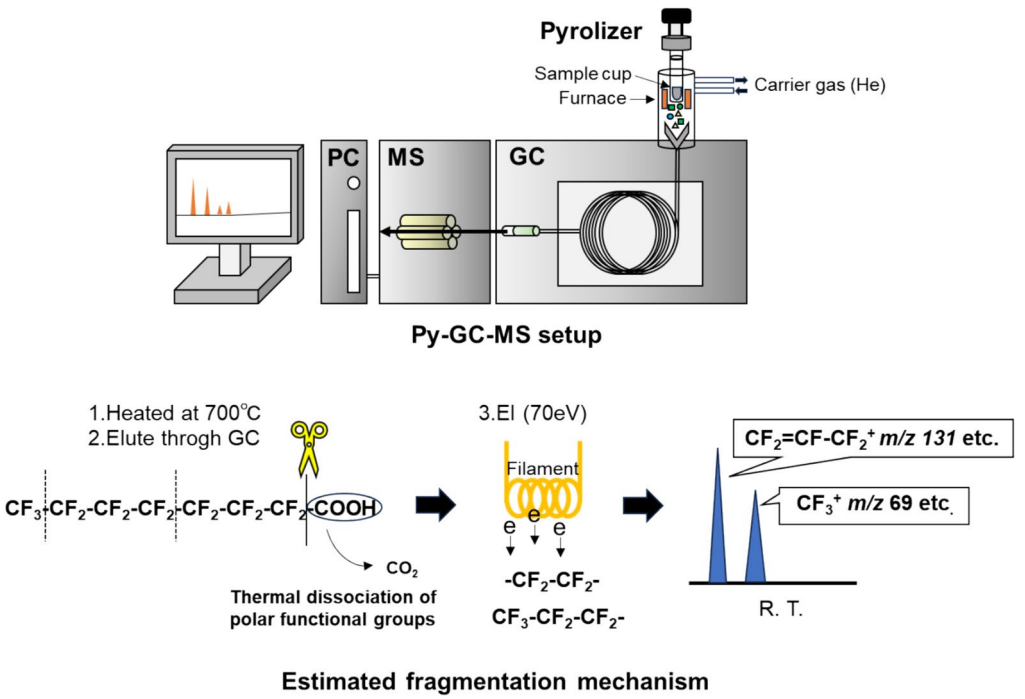


Fig. 1 Schematic of the Py-GC/MS system, illustrating the analytical flow and the estimated mechanism of PFAS fragmentation

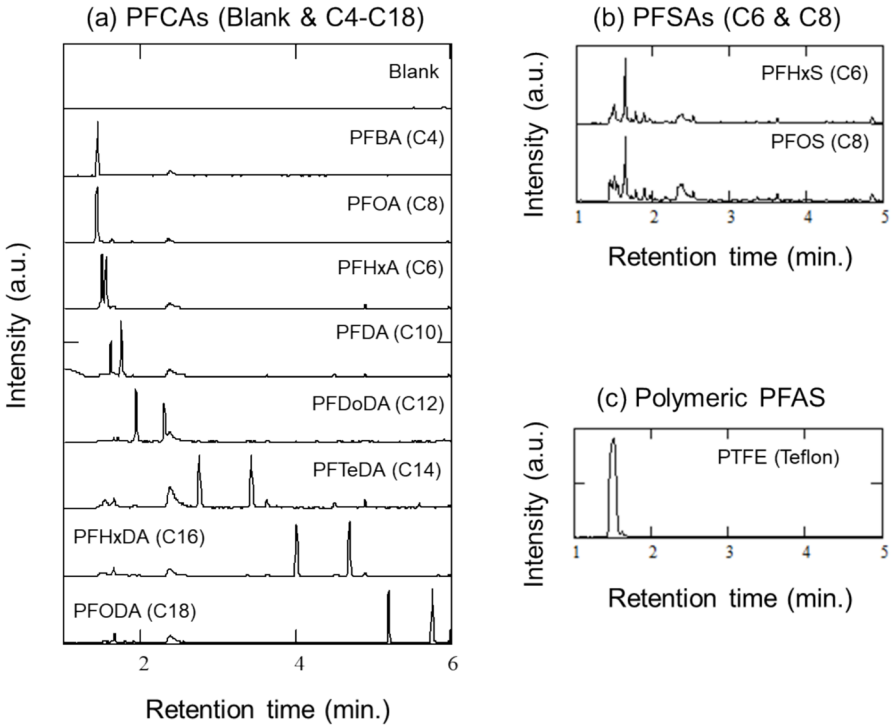


Fig. 2 TIC chromatograms of various non-polymeric PFAS (ea. 1000 mg/kg) and polymeric PFAS: (**a**) C4—C18 PFCAs, (**b**) C6 and C8 PFSAs, and (**c**) PTFE

with scan mode measurements. For reference, the TIC results shown in Fig. 2 have been replaced with the corresponding SIM results and are included as Supplementary Fig. 1. Figure 3 illustrate examples of electron ionization (EI) mass spectra for the TIC chromatographic peaks of a PFOA standard solution (1,000 mg/kg).

The front peak predominantly contained ions representative of perfluoro methylene groups (*m/z* 131), while the rear peak was rich in ions representative of perfluoro methane groups (*m/z* 69). This indicates that the pyrolyzed methylene chains (-CF_2-) elute first, followed by the remaining moieties containing methane groups (CF_3-). The thermally decomposed chain segments lengthen with increasing length of the chain, so the chains are separable in the GC column and peak splitting is observed. Consequently, retention times and peak profiles are useful for estimating perfluoroalkyl chain lengths.

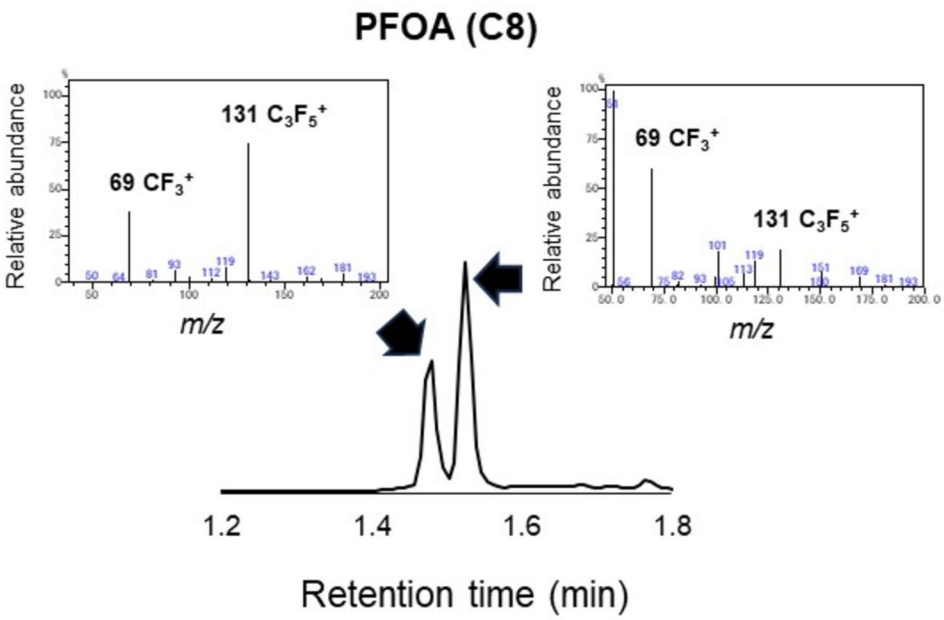


Fig. 3 Examples of Electron Ionization mass spectra for the TIC chromatographic peaks

Furthermore, the Py-GC–MS method enables the detection of polymer-based PFAS samples, which are typically challenging to detect via LC–MS or LC–MS/MS (Fig. 2c). This capability of Py-GC–MS to identify both non-polymer and polymer-based PFAS enhances its effectiveness in universal PFAS screening. In general, longer-chain PFAS compounds contain more perfluoro methylene (-CF_2-) groups in their structures, resulting in stronger peaks than shorter-chain PFAS compounds with fewer -CF_2- groups. Figure 4 compares the peak intensities of the short-chain PFAS compounds (ea. 10 mg/kg) in SIM analysis mode, where "short-chain" refers to PFAS with eight or fewer carbons.

The LOD for PFOA was approximately 1 mg/kg, as evidenced by a t-test (99% confidence) of six repetitive measurements at the 10 mg/kg calibration point ($t = 3.365$). However, as the carbon numbers decrease, the signal at *m/z* 131 diminishes, reducing the signal-to-noise ratio (S/N) and degrading the LOD (Fig. 4). Accordingly, the LODs of the shorter-chain PFAS compounds PFBA and PFHxA increased to 1.5 and 2.2 mg/kg, respectively. Nevertheless, the LODs of the short-chain PFAS compounds remained below 10 mg/kg, equivalent to the 1/100 level of intentional use in consumer products and meeting the intention of comprehensive class regulations on PFAS; that is, to restrict the intentional use of PFAS and encourage the development of safer non-PFAS alternatives. Lower LODs are expected by optimizing GC–MS conditions, including GC column selection, temperature control, split ratio, and sampling amount, with further optimization planned for future studies. Additionally, the linearity of the calibration curve was maintained up to 1,000 mg/kg, with a correlation coefficient of 0.998, indicating that a one-point calibration could suffice for screening purposes (details provided as supplementary Fig. 2).

Following method validation with standard materials, trials for PFAS screening were conducted on several commercial samples. As a critical final step, the test method needed verification with real samples containing diverse classes of PFAS. However, sourcing appropriate PFAS-positive samples proved challenging. Eventually, three real samples were prepared for method validation. Figure 5 illustrates SIM quantitative analyses of these real commercial samples: (a) polyester film, (b) polycarbonate sheet, and (c) polyimide sheet.

The results confirmed the presence of PFAS, correlating roughly with the total fluorine contents obtained by CIC, which ranged from hundreds to thousands of mg F kg^{-1}, i.e. (a) 900 mg F/kg, (b) 1,300 mg F/kg, and (c) 2,000 mg F/kg It is important to note that the SIM settings for PFAS screening by Py-GC–MS specifically target perfluoro methane/methylene groups and do not detect other fluorine sources, such as partially fluorinated compounds. Given that Py-GC–MS primarily targets PFAS containing perfluoro groups, its results may not align with those of the CIC method, which quantifies total fluorine regardless of the source. In fact, the quantitative results of the Py-GC–MS screening were (a) 300 mg/kg, (b) 500 mg/kg, and (c) 980 mg/kg, respectively. This method enables PFAS quantification from GC–MS data in SIM mode without CIC processing. Another demonstration using real samples for comparison with LC–MS/MS results has

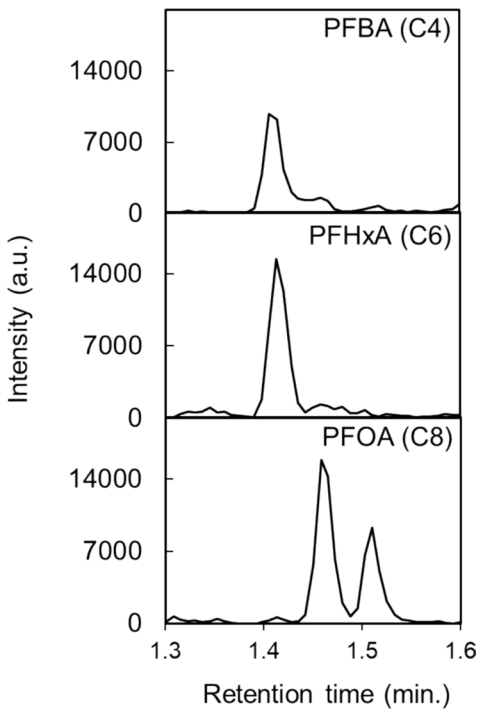


Fig. 4 Comparison of peak intensities in SIM (*m/z* 131) mode analysis for short-chain PFAS compounds (PFBA, PFHxA and PFOA)

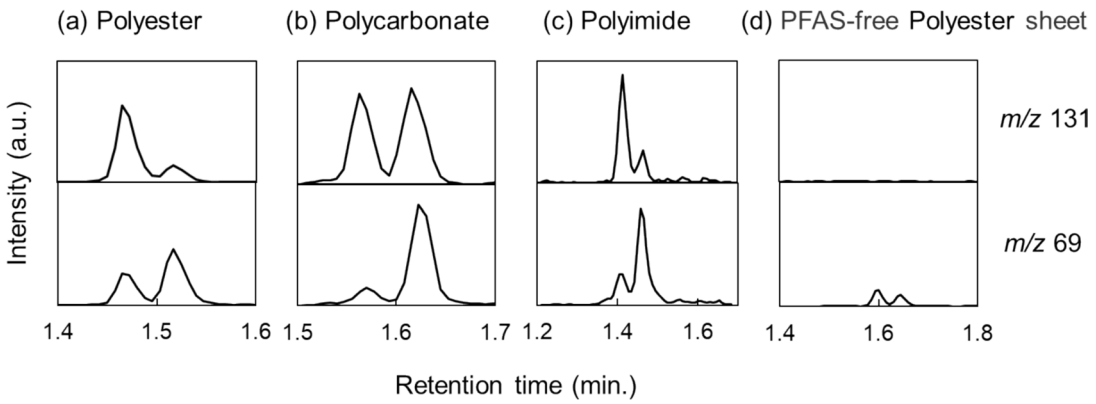


Fig. 5 Examples of SIM quantitative analyses on real commercial samples: (**a**) polyester film, (**b**) polycarbonate sheet, (**c**) polyimide sheet, and (**d**) PFAS-free polyester sheet

been provided as Supplementary Fig. 3. Further verification using other commercially available samples, beyond the scope of this brief communication, remains a future challenge. In addition, this study focuses on the rapid identification of diverse PFAS classes in consumer products not on the precise detection of trace amounts of specific PFAS. In future work, our scope should be broadened to thoroughly identify all forms of PFAS at different locations.

4 Conclusions

In conclusion, the screening method utilizing Py-GC–MS has proven capable of detecting various classes of PFAS, from non-polymeric PFAS of different chain lengths to polymeric PFAS, in under 30 min without the need for complex pre-treatment. Basic validation results included quantitative SIM analysis, confirming an LOD below 1/100 of the typical intentional use concentrations even of short-chain PFAS in consumer products. Therefore, the method detects even low levels of intentionally added PFAS within a sufficient margin. While LC–MS (or LC–MS/MS) remains the preferred method for detecting specific PFAS, such as individual PFOS, PFOA, and their precursors, with very low limits, pyrolysis GC–MS provides a valuable tool for addressing comprehensive class-based regulations through screening approaches. As PFAS regulations evolve from substance bans to comprehensive class-based regulations, the Py-GC–MS screening approach emerges as a promising solution to these regulatory challenges.

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Author contributions S.F. and H.Y. conceived and designed the experiments; H.Y. and S.F. performed the experiments; S.F. and H.Y. analyzed the data; K.O and M.F. reviewed the experimental data; and S.F. and H.Y. wrote the paper.

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Data availability Data will be available upon a reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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